

Template Reflection Individu (comp1)

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1. Kontribusi terhadap kelompok

Saya bertanggung jawab sebagai role 2, yaitu melakukan proses packet capture dan menyediakan bukti (evidence) terkait lalu lintas data jaringan. Kontribusi saya yaitu:

- Melakukan perekaman lalu lintas jaringan (sniffing) secara real-time menggunakan Wireshark saat Role 1 mengeksekusi request API (2 GET dan 2 POST) berbasis format XML.
- Mengisolasi, memfilter, dan menganalisis paket-paket krusial seperti proses TCP 3-Way Handshake serta paket TLS Application Data.
- Menghasilkan output berkas mentah berupa file log jaringan berformat .pcapng dan mendokumentasikan screenshot hasil analisis.

2. Kendala yang dihadapi selama Praktikum

Selama proses pengerjaan praktikum, saya menghadapi beberapa kendala pada saat capturing:

- Pada awal pengujian, saya menerapkan filter berupa keyword http dan xml pada Wireshark, namun visualisasi baris paket yang dicari tidak menghasilkan apa-apa (kosong). Paket yang tertangkap justru didominasi oleh protokol SSDP/UDP milik sistem operasi lokal.

3. Solusi yang Diterapkan

Untuk menyelesaikan kendala tersebut, saya melakukan tindakan troubleshooting sebagai bukti (evidence) dengan langkah-langkah berikut:

- Saya menyadari bahwa endpoint API Beeceptor menggunakan skema HTTPS, yang berarti seluruh muatan HTTP dan tag XML di dalamnya otomatis terenkripsi oleh lapisan Transport Layer Security (TLS).
- Makadari itu, saya mengganti strategi dengan melacak IP server tujuan. Saya menerapkan filter spesifik alamat IP target, yaitu `ip.addr == 159.89.140.122`. Hasilnya, Wireshark berhasil menyaring dan menampilkan urutan paket. Saya berhasil mengidentifikasi proses inisiasi koneksi pada paket pelacakan payload data pada paket berprotokol TLSv1.2 (Application Data).

Capturing on Wi-Fi 2

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ip.addr == 159.89.140.122

No.	Time	Source	Destination	Protocol	Length	Info
55	8.339683700	159.89.140.122	192.168.100.3	TLV1.2	93	Application Data
56	8.339966200	192.168.100.3	159.89.140.122	TCP	54	49382 → 443 [FIN, ACK] Seq=1 Ack=40 Win=253 Len=0
58	8.560821100	159.89.140.122	192.168.100.3	TLV1.2	78	Application Data
59	8.560821700	159.89.140.122	192.168.100.3	TCP	54	443 → 49382 [FIN, ACK] Seq=64 Ack=2 Win=501 Len=0
61	8.560157100	192.168.100.3	159.89.140.122	TCP	54	49382 → 443 [RST, ACK] Seq=2 Ack=64 Win=0 Len=0
71	10.646504100	159.89.140.122	192.168.100.3	TCP	56	443 → 60903 [ACK] Seq=1 Ack=1 Win=501 Len=0
72	10.646551200	192.168.100.3	159.89.140.122	TCP	54	[TCP ACKed unseen segment] 60903 → 443 [ACK] Seq=1 Ack=2 Win=252 Len=0
2198	25.845246800	159.89.140.122	192.168.100.3	TCP	56	[TCP Dup ACK 71#1] 443 → 60903 [ACK] Seq=1 Ack=1 Win=501 Len=0
2199	25.845278200	192.168.100.3	159.89.140.122	TCP	54	[TCP Dup ACK 72#1] 60903 → 443 [ACK] Seq=1 Ack=2 Win=252 Len=0
3124	41.037421200	159.89.140.122	192.168.100.3	TCP	56	[TCP Dup ACK 71#2] 443 → 60903 [ACK] Seq=1 Ack=1 Win=501 Len=0
3125	41.037487900	192.168.100.3	159.89.140.122	TCP	54	[TCP Dup ACK 72#2] 60903 → 443 [ACK] Seq=1 Ack=2 Win=252 Len=0
4412	51.747648300	159.89.140.122	192.168.100.3	TLV1.2	93	[TCP Previous segment not captured], Application Data
4413	51.747963300	192.168.100.3	159.89.140.122	TLV1.2	93	Application Data
4414	51.748158900	192.168.100.3	159.89.140.122	TLV1.2	78	Application Data
4415	51.748267500	192.168.100.3	159.89.140.122	TCP	54	60903 → 443 [FIN, ACK] Seq=64 Ack=41 Win=252 Len=0
4416	51.945228400	159.89.140.122	192.168.100.3	TLV1.2	78	Application Data
4417	51.945229400	159.89.140.122	192.168.100.3	TCP	56	443 → 60903 [ACK] Seq=41 Ack=65 Win=501 Len=0
4418	51.945229900	159.89.140.122	192.168.100.3	TCP	54	443 → 60903 [FIN, ACK] Seq=65 Ack=65 Win=501 Len=0
4419	51.945230000	192.168.100.3	159.89.140.122	TCP	54	60903 → 443 [ACK] Seq=65 Ack=65 Win=501 Len=0

Frame 61: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface 0
Ethernet II, Src: SichuanAILin_97:44:21 (9c:12:21:97:44:21), Dst: HuaweiTechno-13:f5:5b:5b:5b:5b
Internet Protocol Version 4, Src: 192.168.100.3, Dst: 159.89.140.122
Transmission Control Protocol, Src Port: 49382, Dst Port: 443, Seq: 2, Ack: 64, Len: 0

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tcp.port == 443

No.	Time	Source	Destination	Protocol	Length	Info
7846	94.752765900	2606:4700:7::a29f:8...	2001:448a:2082:8e10...	TLV1.2	1152	Application Data
7847	94.752766400	2606:4700:7::a29f:8...	2001:448a:2082:8e10...	TLV1.2	105	Application Data
7848	94.752799500	2001:448a:2082:8e10...	2606:4700:7::a29f:8...	TCP	74	49411 → 443 [ACK] Seq=2956 Ack=1871 Win=255 Len=0
7849	94.780748300	2001:448a:2082:8e10...	2603:1040:5:3::a	TLV1.2	102	Application Data
7850	94.796002200	192.168.100.3	65.1.125.235	TCP	54	49413 → 443 [ACK] Seq=77 Ack=834 Win=251 Len=0
7851	94.801909000	2603:1040:5:3::a	2001:448a:2082:8e10...	TCP	74	443 → 54352 [ACK] Seq=163 Ack=299 Win=624 Len=0
7853	95.025563700	192.168.100.3	159.89.140.122	TCP	66	59820 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
7856	95.182123500	192.168.100.3	108.138.141.100	TCP	55	[TCP Keep-Alive] 59813 → 443 [ACK] Seq=1 Ack=1 Win=509 Len=1
7857	95.187979400	108.138.141.100	192.168.100.3	TCP	66	[TCP Keep-Alive ACK] 443 → 59813 [ACK] Seq=1 Ack=2 Win=169 Len=0 SLE=1 SRE=2
7859	95.219223900	159.89.140.122	192.168.100.3	TCP	66	443 → 59820 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1412 SACK_PERM WS=128
7860	95.219327400	192.168.100.3	159.89.140.122	TCP	54	59820 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
7861	95.219526800	192.168.100.3	159.89.140.122	TLV1.3	571	Client Hello [SNI=mp0eff94b33b385b71a6.free.beeceptor.com]
7862	95.419949300	159.89.140.122	192.168.100.3	TCP	54	443 → 59820 [ACK] Seq=1 Ack=518 Win=64128 Len=0
7863	95.419954200	159.89.140.122	192.168.100.3	TLV1.3	2878	Server Hello, Change Cipher Spec, Application Data
7864	95.419957600	159.89.140.122	192.168.100.3	TLV1.3	1161	Application Data, Application Data, Application Data
7865	95.420039300	192.168.100.3	159.89.140.122	TCP	54	59820 → 443 [ACK] Seq=518 Ack=3932 Win=65280 Len=0
7866	95.423405200	192.168.100.3	159.89.140.122	TLV1.3	118	Change Cipher Spec, Application Data
7867	95.423540600	192.168.100.3	159.89.140.122	TLV1.3	240	Application Data
7871	95.465621100	192.168.100.3	159.89.140.122	TLV1.2	166	Application Data

Frame 7861: Packet, 571 bytes on wire (4568 bits), 571 bytes captured (4568 bits) on interface 0
Ethernet II, Src: SichuanAILin_97:44:21 (9c:12:21:97:44:21), Dst: HuaweiTechno-13:f5:5b:5b:5b:5b
Internet Protocol Version 4, Src: 192.168.100.3, Dst: 159.89.140.122
Transport Layer Security

Capturing from Wi-Fi 2

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dns.qry.name contains beeceptor

No.	Time	Source	Destination	Protocol	Length	Info
7842	94.638219600	2001:448a:2082:8e10...	2001:4489:2082:102::2	DNS	119	Standard query 0x530f AAAA mp0eff94b33b385b71a6.free.beeceptor.com
7843	94.662694300	2001:448a:2082:8e10...	2001:4489:2082:102::2	DNS	119	Standard query 0x530f AAAA mp0eff94b33b385b71a6.free.beeceptor.com
7852	95.020928300	2001:4489:2082:102::2	2001:448a:2082:8e10...	DNS	183	Standard query response 0x530f AAAA mp0eff94b33b385b71a6.free.beeceptor.com SOA ns1.digitalocean.com
7854	95.043606300	2001:4489:2082:102::2	2001:448a:2082:8e10...	DNS	183	Standard query response 0x530f AAAA mp0eff94b33b385b71a6.free.beeceptor.com SOA ns1.digitalocean.com
11426	139.717799800	2001:448a:2082:8e10...	2001:4489:2082:102::2	DNS	97	Standard query 0xa93e HTTPS app.beeceptor.com
11427	139.717951900	2001:448a:2082:8e10...	2001:4489:2082:102::2	DNS	97	Standard query 0x5fbf AAAA app.beeceptor.com
11429	139.718099200	2001:448a:2082:8e10...	2001:4489:2082:102::2	DNS	97	Standard query 0x88d2 A app.beeceptor.com
11505	139.952891800	2001:4489:2082:102::2	2001:448a:2082:8e10...	DNS	161	Standard query response 0xa93e HTTPS app.beeceptor.com SOA ns1.digitalocean.com
11513	140.101088600	2001:448a:2082:8e10...	2001:4489:2082:102::2	DNS	97	Standard query 0x7359 AAAA app.beeceptor.com
11514	140.101266900	2001:448a:2082:8e10...	2001:4489:2082:102::2	DNS	97	Standard query 0x93f6 A app.beeceptor.com
11515	140.167079100	2001:4489:2082:102::2	2001:448a:2082:8e10...	DNS	125	Standard query response 0x5fbf AAAA app.beeceptor.com AAAA 2604:a880:2:d0::2083:1
11518	140.471141500	2001:4489:2082:102::2	2001:448a:2082:8e10...	DNS	125	Standard query response 0x7359 AAAA app.beeceptor.com AAAA 2604:a880:2:d0::2083:1
11519	140.540085700	2001:4489:2082:102::2	2001:448a:2082:8e10...	DNS	113	Standard query response 0x93f6 A app.beeceptor.com A 159.89.140.122
11533	140.904521000	2001:4489:2082:102::2	2001:448a:2082:8e10...	DNS	113	Standard query response 0x88d2 A app.beeceptor.com A 159.89.140.122
31500	331.639426000	2001:448a:2082:8e10...	2001:4489:2082:102::2	DNS	119	Standard query 0x7feb AAAA mp0eff94b33b385b71a6.free.beeceptor.com
31501	331.659679700	2001:448a:2082:8e10...	2001:4489:2082:102::2	DNS	119	Standard query 0x7feb AAAA mp0eff94b33b385b71a6.free.beeceptor.com
31519	332.603118100	2001:4489:2082:102::2	2001:448a:2082:8e10...	DNS	183	Standard query response 0x7feb AAAA mp0eff94b33b385b71a6.free.beeceptor.com SOA ns1.digitalocean.com
31521	332.809092200	2001:4489:2082:102::2	2001:448a:2082:8e10...	DNS	183	Standard query response 0x7feb AAAA mp0eff94b33b385b71a6.free.beeceptor.com SOA ns1.digitalocean.com

Frame 7854: Packet, 183 bytes on wire (1464 bits), 183 bytes captured (1464 bits) on Ethernet II, Src: HuaweiTechno_13:f5:98 (94:d5:4d:13:f5:98), Dst: SichuanAILin_97:44 (02:00:00:00:00:02) Internet Protocol Version 6, Src: 2001:4489:2082:102::2, Dst: 2001:448a:2082:8e10:b09 User Datagram Protocol, Src Port: 53, Dst Port: 63023 Domain Name System (response) Transaction ID: 0x530f Flags: 0x8180 Standard query response, No error Questions: 1 Answer RRs: 0 Authority RRs: 1 Additional RRs: 0 Queries: Authoritative name servers

Capturing from Wi-Fi 2

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tcp.flags.syn == 1

No.	Time	Source	Destination	Protocol	Length	Info
7853	95.025563700	192.168.100.3	159.89.140.122	TCP	66	59820 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
7859	95.219223900	159.89.140.122	192.168.100.3	TCP	66	443 → 59820 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1412 SACK_PERM WS=128
7912	95.862372100	192.168.100.3	104.18.33.150	TCP	66	59821 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
7913	95.882556700	104.18.33.150	192.168.100.3	TCP	66	443 → 59821 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1400 SACK_PERM WS=8192
7996	96.406302300	192.168.100.3	35.206.35.210	TCP	66	58974 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
8011	96.631943600	35.206.35.210	192.168.100.3	TCP	66	443 → 58974 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1412 SACK_PERM WS=128
8829	100.983878400	192.168.100.3	52.2.123.41	TCP	66	58975 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
8857	101.227481400	52.2.123.41	192.168.100.3	TCP	66	443 → 58975 [SYN, ACK] Seq=0 Ack=1 Win=26883 Len=0 MSS=1412 SACK_PERM WS=256
9919	120.682645500	192.168.100.3	52.168.117.170	TCP	66	58977 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
9920	120.683209500	192.168.100.3	52.168.117.170	TCP	66	58978 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
9921	120.943393000	52.168.117.170	192.168.100.3	TCP	66	443 → 58977 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 WS=256 SACK_PERM
9922	120.943393800	52.168.117.170	192.168.100.3	TCP	66	443 → 58978 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 WS=256 SACK_PERM
10112	122.062695400	2001:448a:2082:8e10...	2606:4700:4401::ac4	TCP	86	55799 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1432 WS=256 SACK_PERM
10113	122.084377000	2606:4700:4401::ac4	2001:448a:2082:8e10...	TCP	86	443 → 55799 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1360 SACK_PERM WS=8192
11136	136.528462500	2001:448a:2082:8e10...	2001:4488:f931::766	TCP	86	59742 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1432 WS=256 SACK_PERM
11137	136.534016400	2001:4488:f931::766	2001:448a:2082:8e10...	TCP	86	443 → 59742 [SYN, ACK] Seq=0 Ack=1 Win=64800 Len=0 MSS=1344 SACK_PERM WS=128
11520	140.541071000	2001:448a:2082:8e10...	2604:a880:2:d0::208	TCP	86	53129 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1432 WS=256 SACK_PERM
11521	140.541209200	2001:448a:2082:8e10...	2604:a880:2:d0::208	TCP	86	59653 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1432 WS=256 SACK_PERM
11532	140.702856700	2001:448a:2082:8e10...	2604:a880:2:d0::208	TCP	96	60706 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1432 WS=256 SACK_PERM

Frame 7853: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface Ethernet II, Src: SichuanAILin_97:44:21 (9c:12:21:97:44:21), Dst: HuaweiTechno_13:f5:98 Internet Protocol Version 4, Src: 192.168.100.3, Dst: 159.89.140.122 Transmission Control Protocol, Src Port: 59820, Dst Port: 443, Seq: 0, Len: 0